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# Step 1: Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import OrdinalEncoder
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Step 2: Mount Drive
from google.colab import drive
drive.mount('/content/drive')

file_path = '/content/drive/MyDrive/Iris Dataset - Public Livelihood Data.csv'
df = pd.read_csv(file_path)

# Step 3: Prepare Features and Target
X = df.drop(columns=['Salary'])
y = df['Salary']

# Step 4: Encode Categorical Features
encoder = OrdinalEncoder()
X_encoded = pd.DataFrame(encoder.fit_transform(X), columns=X.columns)

# Step 5: Split the Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X_encoded, y, test_size=0.2, random_state=42
)

print("\nTraining set size:", X_train.shape)
print("Testing set size: ", X_test.shape)

# Step 6: Train the Naive Bayes Classifier
model = CategoricalNB()
model.fit(X_train, y_train)

# Step 7: Make Predictions
y_pred = model.predict(X_test)

# Step 8: Evaluate the Classifier
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", round(accuracy, 4))

print("\nPredicted labels (first 20):", y_pred[:20])
print("Actual labels (first 20):", y_test.values[:20])

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

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Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)

Training set size: (26048, 7)
Testing set size: (6513, 7)

Accuracy: 0.8257

Predicted labels (first 20): ['<=50K' '<=50K' '>50K' '<=50K' '<=50K' '>50K' '>50K' '<=50K' '<=50K'
'<=50K' '<=50K' '<=50K' '<=50K' '<=50K' '<=50K' '<=50K' '<=50K' '>50K' '>50K'
'<=50K' '<=50K']

Actual labels (first 20): ['<=50K' '<=50K' '>50K' '<=50K' '<=50K' '>50K' '>50K' '<=50K' '<=50K'
'>50K' '<=50K' '<=50K' '>50K' '<=50K' '<=50K' '<=50K' '>50K' '<=50K'
'>50K' '<=50K']

Classification Report:

	precision	recall	f1-score	support
<=50K	0.87	0.90	0.89	4942
>50K	0.66	0.58	0.62	1571
accuracy			0.83	6513
macro avg	0.76	0.74	0.75	6513
weighted avg	0.82	0.83	0.82	6513

Confusion Matrix:

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[[4460 482]  
 [ 653 918]]
```